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DiffMat: Data-driven inverse design of energy-absorbing metamaterials using diffusion model

Haoyu Wang [a], Zongliang Du [a,b], Fuyong Feng [c,d], Zhong Kang [e], Shan Tang [a,b,*], Xu Guo [a,b,**]

[a] State Key Laboratory of Structural Analysis, Optimization and CAE Software for Industrial Equipment, Department of Engineering Mechanics, Dalian University of Technology, Dalian, 116023, PR China
[b] Ningbo Institute of Dalian University of Technology, Ningbo, 315016, PR China
[c] China North Artificial Intelligence & Innovation Research Institute, Beijing, 100072, PR China
[d] Collective Intelligence & Collaboration Laboratory, Beijing, 100072, PR China
[e] China North Vehicle Research Institute, Beijing, 100072, PR China

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ABSTRACT

Energy-absorbing materials and structures are widely applied in industrial areas. Presently, design methods of energy-absorbing metamaterials mainly rely on empirical or bio-inspired configurations. Inspired by AI-generated content, this paper proposes a novel inverse design framework for energy-absorbing metamaterial using diffusion model called DiffMat, which can be customized to generate microstructures given desired stress–strain curves. DiffMat learns the conditional distribution of microstructure given mechanical properties and can realize the one-to-many mapping from properties to geometries. Numerical simulations and experimental validations demonstrate the capability of DiffMat to generate a diverse array of microstructures based on given mechanical properties. This indicates the validity and high accuracy of DiffMat in generating metamaterials that meet the desired mechanical properties. The successful demonstration of the proposed inverse design framework highlights its potential to revolutionize the development of energy-absorbing metamaterials and underscores the broader impact of integrating AI-inspired methodologies into metamaterial design and engineering.

1. Introduction

Energy-absorbing materials and structures are extensively utilized in aerospace, transportation, and human protection applications due to their exceptional impact mitigation properties [1]. Classical energy-absorbing structures, including sandwich structures [2], plate structures [3], honeycombs [4], and foams [5], have been extensively studied in previous research [3]. In order to enhance the energy absorption capacity, various designs have been proposed to improve classical structures. For instance, composite [6] materials are used for reinforcement and utilizing fold lines of origami structures as plastic hinges for energy absorption improvement [7]. Bio-inspired design is another approach to obtain structures with energy-absorbing properties by mimicking the internal structures found in living organisms [8]. For example, tubular nested geometries inspired by Haversian bone architecture are designed to develop high-efficiency energy-absorbing structures [9], and concentric hexagonal energy-absorbing

* Corresponding author at: State Key Laboratory of Structural Analysis, Optimization and CAE Software for Industrial Equipment, Department of Engineering Mechanics, Dalian University of Technology, Dalian, 116023, PR China.
** Corresponding author.
E-mail addresses: shantang@dlut.edu.cn (S. Tang), guoxu@dlut.edu.cn (X. Guo).

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panels are designed based on the distribution characteristics of bones and bone marrow [10]. These designs are realized through empirical or bio-inspired configurations, adjusting geometric parameters and material types to achieve different energy absorption effects. Collectively, they are referred to as the forward design method for energy-absorbing structures.

Crashworthy designers have taken into account multiple criteria when aiming to design an ideal energy absorber, encompassing not only high energy absorption performance but also considerations of stability and safety [11]. Specific energy absorption (SEA) and specific energy absorption per unit volume (SEA_v) are commonly used to evaluate the energy absorption performance of these structures [8]. Low initial peak crushing stress (IPCS), stable platform stress (σ_m), and high densification strain (ϵ_m) are key indicators of adequate energy-absorbing protection [12]. An ideal energy-absorbing structure should exhibit a load–displacement (stress–strain) curve with a rectangular or trapezoidal shape, which ensures efficient energy absorption [13]. By setting the desired energy absorption curve as the design goal and determining the base material, the only remaining variable is the material distribution within the structure. Designers can tailor the behavior of the metamaterial and enhance energy absorption by carefully manipulating the material distribution to meet specific design objectives and performance criteria.

Topology optimization-based design methods [14] directly satisfy design objectives without strict material requirements, simplifying the design process and improving design efficiency. It is now a mature field with various methods, including the SIMP method [15,16], level set method [17], evolutionary structural optimization method [18], and moving morphable components methods [19,20]. These methods require rigorous sensitivity derivations and exhibit rapid convergence rates based on finite element (FE) analysis. Existing research on topological optimization of mechanical metamaterials has primarily focused on objectives such as minimum compliance [21], negative Poisson's ratio [22], multiscale optimization [23,24] and energy dissipation [25]. Since topology optimization of energy-absorbing metamaterials must account for geometric, material, and contact nonlinearities [12], gradient-based topology optimization methods face significant challenges in energy-absorbing metamaterial design. Meanwhile, intelligent algorithm-based design methods have shown promising performance in addressing complex multiobjective [26], non-convex, or time-varying optimization problems that are challenging for sensitivity-based methods. To address the issue of energy absorption, Zeng et al. [12] proposed a genetic algorithm-based inverse design optimization method for energy-absorbing metamaterials, in which a series of structural mutation algorithms and design-domain-independent mesh generation method are developed to improve the efficiency of finite element analysis and optimization iteration. The highly flexible model representation of intelligent algorithms leads to improved convergence properties. However, intelligent algorithms suffer from lower optimization efficiency than gradient-based algorithms due to the lack of sensitivity guidance, particularly when dealing with strong nonlinear behaviors.

With the rapid development of artificial intelligence (AI) technology, the generation of AI-generated content (AIGC) has garnered significant attention beyond the field of computer science. AIGC refers to content generated using advanced AI generation techniques (GAI) rather than being created by human authors, enabling the automated creation of large amounts of content within a short period [27], such as ChatGPT [28]. Commonly used deep generative models for AIGC include autoregressive models (ARM) [29–31], variational autoencoders (VAE) [32], generative adversarial networks (GAN) [33,34], flow-based models [35,36], and denoising diffusion probabilistic models (DDPM) [37].

In the region of material design, the progress of data-driven paradigms has significantly expedited the exploration of inverse materials design methodologies. From the perspective of data representation, data-driven approaches for the inverse design of mechanical materials can be categorized into two groups. The first category uses parameterized models to represent materials, with building surrogate models to explore the design parameters and achieve desired objectives [38]. Wang et al. [39] introduced an innovative approach that combines machine learning (ML) with genetic algorithms (GA) for the inverse design of shell-based mechanical metamaterials. Two strategies are employed to develop artificial neural networks (ANNs) capable of predicting the loading curves of these metamaterials under compression. Subsequently, GA is used to design configurations that achieve specific, customized loading curves. Zhou et al. [40] designed and optimized the anti-tetrachiral structure by integrating a deep learning model with GA. Within this category, material structures are typically transformed into a collection of topological parameters, allowing for the effective exploration of design parameters through the optimization capabilities of machine learning algorithms. The second category of data-driven design methods focuses on designing within a high-dimensional topological space rather than designing topological parameters directly. A common strategy entails partitioning the topological space into a series of pixels. Generative models [32,33] are commonly employed to generate a pixelized representation of the high-dimensional topological matrix. Generative models are utilized to generate high-dimensional topological structures represented by pixels. Typically, the desired material properties are used as conditions to guide the generative models, ensuring that the trained models can generate materials corresponding to the target material properties. The advantage of the second category lies in its compatibility with complex implicit topologies. The resulting design is represented by discrete pixels and can be further transformed into additive manufacturing through a smoothing process. This category of design approaches has been successfully applied in the design of mechanics metamaterials [41,42]. Kollmann et al. [43] developed a deep learning model based on a convolutional neural network (CNN) that predicts optimal metamaterial designs, which can non-iteratively optimize metamaterials for either maximizing the bulk modulus, maximizing the shear modulus, or minimizing the Poisson's ratio. Wang et al. [44] proposed an inverse homogenization generative adversarial network (IH-GAN) to accurately model the property to geometry mapping using a conditional deep generative model and reduce the complexity of IH-GAN by using the implicit function parameterization to represent unit cell geometries. Vlassis and Sun [45] introduce a denoising diffusion algorithm to identify microstructures with finely-tuned nonlinear properties, which can effectively generate microstructures with these properties within the latent space of the training data. Bastek and Kochmann [46] proposed video diffusion generative models for inverse design of nonlinear mechanical metamaterials, which trained on full-field data of periodic stochastic cellular structures can successfully predict and tune their nonlinear deformation and stress response under compression in the large-strain region, which breaks from the common strategy of directly learning a map from property to design

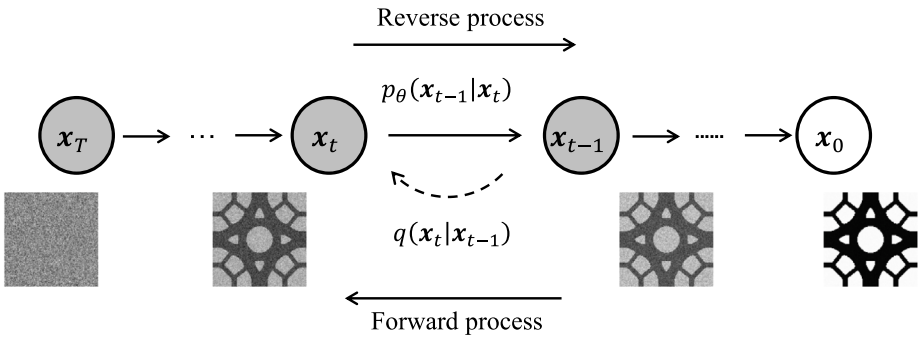


Fig. 1. Diagram of denoising diffusion probabilistic model.

and to extend the framework to intrinsically estimate the expected deformation path and the full-field internal stress distribution. However, the aforementioned work has not focused on the AI-generative design for energy-absorbing metamaterials. Additionally, there has been little experimental validation of these methods, limiting their reliability and effectiveness in practical applications.

This paper presents a data-driven inverse design framework for energy-absorbing metamaterial using diffusion model, namely **DiffMat**, which can generate microstructures based on desired stress–strain curves. The core of this framework lies in the diffusion model, which is adapted to learn the conditional distribution of microstructures corresponding to given stress–strain curves. This capability allows DiffMat to perform a one-to-many mapping, transforming desired mechanical properties into a diverse set of potential geometrical configurations. This framework ensures the accuracy and efficiency of energy-absorbing metamaterial design while highlighting the broader impact of integrating AI-inspired methodologies into material design and engineering.

2. Methodology

In this section, we first introduce the theory of denoising diffusion probabilistic model (DDPM). Subsequently, we extend the DDPM to design energy-absorbing metamaterials and present a novel framework called DiffMat, capable of generating microstructures based on desired stress–strain curves.

2.1. Denoising diffusion probabilistic model

Given samples from a data distribution $q(\boldsymbol{x}_0)$, DDPM [37] is unconditional generative models aiming to learn a model distribution $p_\theta(\boldsymbol{x}_0)$ that approximates $q(\boldsymbol{x}_0)$. The diffusion process consists of a total of T steps. Let $\boldsymbol{x}_t$ for $t = 1, \dots, T$ be a sequence of latent variables from the same sample space of $\boldsymbol{x}_0 \in \mathbf{X}$. DDPM are latent variable models of the form $p_\theta(\boldsymbol{x}_0) = \int p_\theta(\boldsymbol{x}_{0:T})\mathrm{d}\boldsymbol{x}_{1:T}$. It contains two processes, namely the forward process and the reverse process, as shown in Fig. 1.

2.1.1. Forward process

The forward process is defined by a Markov chain which progressively adds Gaussian noise to the observation $\boldsymbol{x}_0$

$$q\left(\boldsymbol{x}_{1:T} \mid \boldsymbol{x}_0\right) = \prod_{t=1}^{T} q\left(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}\right) \tag{1}$$

where $q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1})$ is a Gaussian distribution as

$$q\left(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}\right) = \mathcal{N}\left(\boldsymbol{x}_t; \sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t \mathbf{I}\right) \tag{2}$$

and $\beta_t \in (0,1)$ that represents the noise level at forward step t and $\beta_1 < \beta_2 < \cdots < \beta_T$ is an increasing variance schedule. Let $\alpha_t = 1-\beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, a special property of the forward process is that the distribution of $\boldsymbol{x}_t$ given $\boldsymbol{x}_0$ has a close form:

$$q\left(\boldsymbol{x}_t \mid \boldsymbol{x}_0\right) = \mathcal{N}\left(\boldsymbol{x}_t; \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0, \left(1-\bar{\alpha}_t\right)\mathbf{I}\right) \tag{3}$$

which can also be expressed through the reparameterization trick [32]

$$\boldsymbol{x}_t = \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon} \tag{4}$$

where $\boldsymbol{\epsilon} \in \mathcal{N}(\mathbf{0}, \mathbf{I})$ as a sampled noise. The above property allows us to directly sample $\boldsymbol{x}_t$ at any arbitrary noise level t, instead of computing the forward process step by step.

2.1.2. Reverse process

The reverse process denoises $\boldsymbol{x}_T$ to recover $\boldsymbol{x}_0$ recurrently. It also follows a Markov chain but with learnable Gaussian transitions starting with $p(\boldsymbol{x}_T) = \mathcal{N}(\boldsymbol{x}_T; \mathbf{0}, \mathbf{I})$, which is defined as

$$p_\theta\left(\boldsymbol{x}_{0:T}\right) = p\left(\boldsymbol{x}_T\right) \prod_{t=1}^{T} p_\theta\left(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t\right) \tag{5}$$

which

$$p_\theta\left(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t\right) = \mathcal{N}\left(\boldsymbol{x}_{t-1}; \boldsymbol{\mu}_\theta\left(\boldsymbol{x}_t, t\right), \boldsymbol{\sigma}_\theta\left(\boldsymbol{x}_t, t\right) \mathbf{I}\right) \tag{6}$$

with shared and learnable parameters $\boldsymbol{\theta}$. Here we choose the same parameterization of $p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)$ as in [37] given its encouraging performance in image generation:

$$\mu_\theta(\boldsymbol{x}_t, t) = \frac{1}{\bar{\alpha}_t}\left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)\right) \tag{7}$$

$$\sigma_\theta(\boldsymbol{x}_t, t) = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t \tag{8}$$

where $\boldsymbol{\epsilon}_\theta$ is a trainable function that decides the amount of noise should be removed during the current denoising step. The parameters $\boldsymbol{\theta}$ are learned by solving the following optimization problem:

$$\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) = \min_{\boldsymbol{\theta}} \mathbb{E}_{\boldsymbol{x}_0 \sim q(\boldsymbol{x}_0), \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), t} \| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\boldsymbol{x}_t, t\right) \|_2^2 \tag{9}$$

Since we already know $\boldsymbol{x}_0$ in the training stage and recall Eq. (4), the above training objective of unconditional generation can be specified as

$$\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) = \min_{\boldsymbol{\theta}} \mathbb{E}\left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\sqrt{\bar{\alpha}_t}\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t\right)\right\|_2^2 \tag{10}$$

More derivation details can be found in [37]. Once the training phase of the diffusion model is successfully completed, we can sample $\boldsymbol{x}_0$ by Eq. (5) starting from the Gaussian noise $\boldsymbol{x}_T$. This reverse process resembles Langevin dynamics, where we initially sample from the most noise-perturbed distribution and gradually decrease the noise scale step by step until reaching the smallest one [47]. Further details can be found in the original paper [37].

2.1.3. Conditional diffusion model

From a methodological perspective, there exist two approaches to conditional control generation, namely classifier-guidance [48] and classifier-free scheme [49,50]. The former obviates the necessity for retraining the diffusion model and enables straightforward control at a minimal expense, while the latter necessitates the retraining of the diffusion model, incurring higher costs but allowing for finer control. In the subsequent sections, we present the classifier-free scheme, which reverse diffusion process is characterized by its simplicity and direct definition as follows:

$$p_\theta\left(\boldsymbol{x}_{0:T} \mid \boldsymbol{y}\right) = p\left(\boldsymbol{x}_T\right) \prod_{t=1}^{T} p_\theta\left(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{y}\right) \tag{11}$$

which

$$p_\theta\left(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{y}\right) = \mathcal{N}\left(\boldsymbol{x}_{t-1}; \boldsymbol{\mu}_\theta\left(\boldsymbol{x}_t, \boldsymbol{y}, t\right), \boldsymbol{\sigma}_\theta\left(\boldsymbol{x}_t, \boldsymbol{y}, t\right)\right) \tag{12}$$

which $\boldsymbol{y}$ represents the condition. Furthermore, the training objective in Eq. (9) can be rewritten as a conditional one:

$$\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) = \min_{\boldsymbol{\theta}} \mathbb{E}_{\boldsymbol{x}_0, \boldsymbol{y}, \boldsymbol{\epsilon}} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\boldsymbol{x}_t, t | \boldsymbol{y}\right)\right\|_2^2 \tag{13}$$

2.2. DiffMat

The objective of metamaterial design is to generate metamaterial topology structure $\boldsymbol{x}^\rho$ given the stress–strain information $\boldsymbol{y}^\sigma$, which is formulated as:

$$\mathcal{F} : \boldsymbol{y}^\sigma \rightarrow \boldsymbol{x}^\rho \tag{14}$$

where $\boldsymbol{y}^\sigma \in \mathbb{R}^k$ denote k-dimensional stress–strain information and $\boldsymbol{x}^\rho \in [0,1]^{N \times N}$ denote microstructural topology. The Eq. (11) can be rewritten as

$$p_\theta\left(\boldsymbol{x}^\rho_{0:T} \mid \boldsymbol{y}^\sigma\right) = p\left(\boldsymbol{x}^\rho_T\right) \prod_{t=1}^{T} p_\theta\left(\boldsymbol{x}^\rho_{t-1} \mid \boldsymbol{x}^\rho_t, \boldsymbol{y}^\sigma\right) \tag{15}$$

And Eq. (13) can be rewritten as

$$\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) = \min_{\boldsymbol{\theta}} \mathbb{E}_{\boldsymbol{x}_0, \boldsymbol{y}^\sigma, \boldsymbol{\epsilon}} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\boldsymbol{x}^\rho_t, t | \boldsymbol{y}^\sigma\right)\right\|_2^2 \tag{16}$$

Training. In the training process, given the stress–strain information $\boldsymbol{y}^{\sigma}$ all, we sample noise targets by Eq. (4), and then train ϵ_{θ} by the loss function in Eq. (16). The training procedure of DiffMat is presented in Algorithm 1.

Algorithm 1 Diffusion model training of DiffMat

Input: metamaterial topology structure $\boldsymbol{x}_0^{\rho}$, stress-strain curve $\boldsymbol{y}^{\sigma}$, number of diffusion step T, variance schedule $\{\beta_1, \cdots, \beta_T\}$
Output: Trained denoising function ϵ_{θ}

```
while converged do
    Sample t ~ Uniform(1, ..., T)
    Sample ε ~ N(0, I) where the dimension of ε corresponds to x_0^ρ
    Calculate noisy targets x_t^ρ by Eq. (4)
    Calculate L(θ) by Eq. (16)
    Take gradient step ∇_θ ||ε − ε_θ(x_t^ρ, t|y^σ)||_2^2
    Update ε_θ using gradient descent method
end while
return ε_θ
```

Sampling. As outlined in Algorithm 2, the inference process utilizes the trained denoising function ϵ_{θ} to sample $\boldsymbol{x}^{\rho}$ step by step according to Eq. (15), under the guidance of $\boldsymbol{y}^{\sigma}$.

Algorithm 2 Diffusion model sampling of DiffMat

Input: stress-strain curve $\boldsymbol{y}^{\sigma}$ number of diffusion step T, trained denoising function ϵ_{θ}
Output: metamaterial topology structure $\boldsymbol{x}_0^{\rho}$

```
Sample initial noise ε ~ N(0, I) where the dimension of ε corresponds to x_0^ρ
for i = T to 1 do
    Sample x_{t-1}^ρ using Eq. (15)
end for
return x^ρ
```

3. Data prepare and model architecture

3.1. Metamaterials generation by homogenization-based topology optimization

This section introduces the process of generating metamaterials topology dataset for subsequent FE simulations. In this paper, we used the Matlab code [51], which serves as a tool for generating metamaterials dataset. The homogenization-based topology optimization problem can be mathematically formulated as follows:

$$\begin{aligned} &\text{find}: \boldsymbol{\rho} = \begin{bmatrix} \rho_1 & \rho_2 & \cdots & \rho_N \end{bmatrix} \\ &\text{Maximize}: J(\boldsymbol{u}, \boldsymbol{\rho}) = \sum_{i,j,k,l=1}^{d} f(E_{ijkl}^{H}(\boldsymbol{\rho})) \\ &\text{Subject to}: \begin{cases} \boldsymbol{K}(\boldsymbol{\rho})\boldsymbol{u} = \boldsymbol{f} \\ V = \sum_{e=1}^{N} \rho V_e \leqslant V_{max} \\ 0 \leq \rho_e \leq 1, e = 1, \ldots, N \end{cases} \end{aligned} \tag{17}$$

where $\boldsymbol{\rho}$ denotes the vector of design variables. J is the objective function defined by a combination of the homogenized elastic tensor E^H. $\boldsymbol{K}$ is the global stiffness matrix of the physical problem obtained from the FE discretization. $\boldsymbol{f}$ is the external force vector. $\boldsymbol{u}$ is the displacement vector which satisfies the periodic boundary condition (PBC). V is the volume constraint with an upper bound of V_{max}. And the Young's modulus E of the artificial material is expressed as a power-law interpolation function of the mapped material field [15], namely:

$$E(\rho) = E_{min} + \rho^{p}(E_0 - E_{min}) \tag{18}$$

where E_0 is the Young's modulus of the fully solid material, p is the penalization factor, and E_{min} is a small positive value required to avoid stiffness singularity in the FE analysis.

Fig. 2 shows the metamaterial generation process. Through topology optimization, 30,000 metamaterials have been systematically generated with volume fractions ranging from 28% to 52% by changing the initial guess of material topology layout, filter radius, and penalization factor. Both horizontal and vertical dimensions consist of 128 design variables. In order to ensure a clear boundary, we used a large penalty factor such that $p \geq 3$ and then binarized the topological microstructure. Fig. 3 visually represents the binarized metamaterials with different volume fraction.

After obtaining binary microstructures, the jagged nature of the edges poses a challenge when directly applied in FE simulation, potentially causing significant mesh distortions and consequently impacting the convergence and precision of the simulations. As

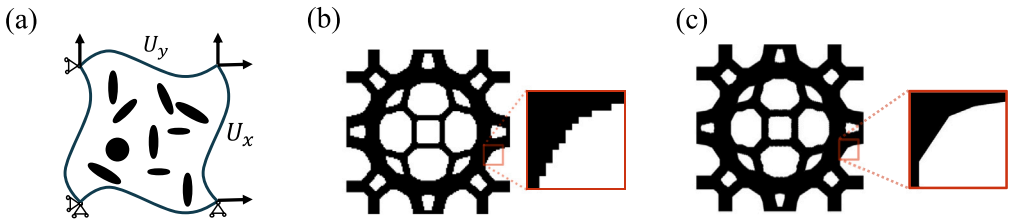


Fig. 2. Metamaterial generation process. (a) periodic boundary conditions; (b) microstructure generation by topology optimization; (c) density contour method.

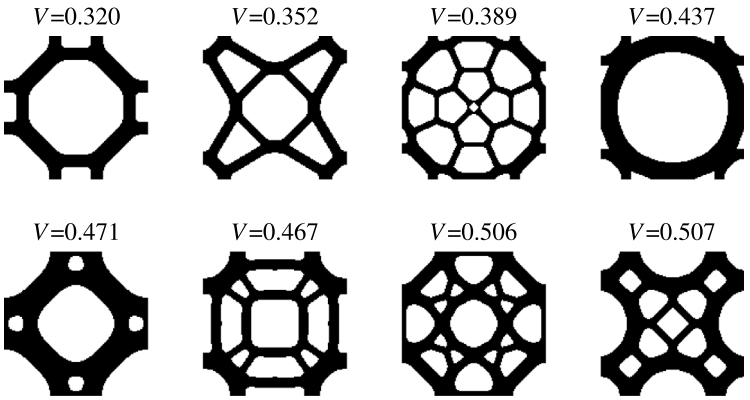


Fig. 3. Microstructures generated by topology optimization.

shown in Fig. 2, we employed the density contour approach for boundary detection, in which boundaries are extracted using the isoline of the density fields [52,53]. Then, the Ramer–Douglas–Peucker (RDP) [54,55] algorithm effectively reduces the detected boundary data points. Such a processing approach is crucial for achieving reliable simulation results, particularly in the modeling and simulating complex geometry.

3.2. Elasto-plastic model parameter calibration

In order to obtain the mechanical properties of 304 stainless steel (304SS) material, the tensile specimen used in the experiment was designed according to GB/T 228.1-2021 [56]. And its drawing is shown in Fig. 4(a), and the unit is in mm. The experiment is conducted using a tensile testing machine, and the displacement speed of the collet is set to 2 mm/min (strain rate 6.67×10^{-4} s^{-1}) to conform to the quasi-static (strain rate roughly between 10^{-5} s^{-1} and 10^{-1} s^{-1}) tensile process. The installation method is shown in Fig. 4(b) to ensure the accuracy of the force–displacement data of 304SS obtained during the stretching process. A high-precision extensometer calibrator collects the axial displacement of the specimen during the stretching process. The force–displacement data of the standard tensile specimen obtained from the tensile test were processed to obtain the true stress–strain curve of the tensile specimen, shown as the blue line in Fig. 4(c).

Assuming small elastic and finite plastic deformations, the deformation rate tensor D_{ij} is regarded as being decomposed into elastic and plastic components [57,58].

$$D_{ij} = D^{\mathrm{e}}_{ij} + D^{\mathrm{p}}_{ij} \tag{19}$$

where the superscripts e and p indicate the elastic and plastic components, respectively. The constitutive relationship between the elastic deformation rate tensor and the Cauchy stress can be formulated as

$$D^{\mathrm{e}}_{ij} = \frac{1+\nu}{E}\sigma^{\nabla J}_{ij} - \frac{\nu}{E}\mathrm{tr}(\sigma^{\nabla J}_{ij})I_{ij} \tag{20}$$

where E represents Young's modulus and ν denotes Poisson's ratio. The term $\sigma^{\nabla J}_{ij}$ refers to the Jaumann rate of the Cauchy stress, while I_{ij} is the second-order identity tensor. Considering that 304SS is an isotropic material, the elastic modulus E of 304SS can be determined to be approximately 185 GPa by calculating the slope of the curve in the elastic phase, shown in Fig. 4. And Poisson's ratio is selected as 0.3. In this study, the J2 plasticity model with isotropic hardening is used as follows

$$f(\sigma_{ij}, \sigma_y) = \sqrt{\frac{3}{2}s_{ij}s_{ij}} - \sigma_y \tag{21}$$

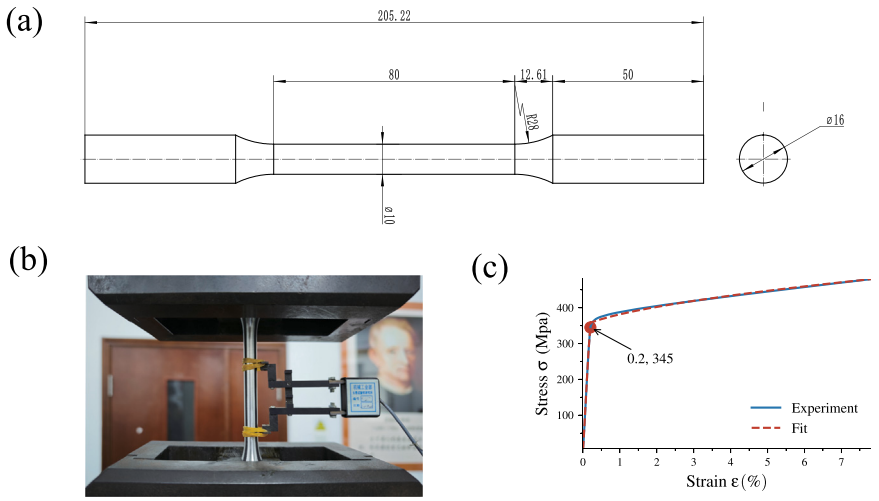


Fig. 4. Tensile experiment: (a) specimen drawing; (b) experimental loading condition; (c) true stress–strain curve.

Table 1
Material parameters for simulation of 306 stainless steel.

Density (tonne/mm^3)	E (GPa)	ν	A (MPa)	B (MPa)	n
7.85×10^{-9}	185	0.3	344	654	0.61

where $s_{ij} = \sigma_{ij} - (1/3)\sigma_{kk}\delta_{ij}$ is the deviatoric stress tensor, and σ_y is yield stress. For associated flow rules that choose the yield function as the plastic potential, one has

$$D^{\mathrm{p}}_{ij} = \dot{\lambda}\frac{\partial f}{\partial \sigma_{ij}} = \dot{\lambda}\sqrt{\frac{3}{2}}\frac{s_{ij}}{\| s_{ij} \|} \tag{22}$$

where $\dot{\lambda} = \sqrt{\frac{2}{3}D^{\mathrm{p}}_{ij}D^{\mathrm{p}}_{ij}}$ represents the rate of effective plastic strain, which corresponds to the strain rate in the case of small deformations. Based on the Johnson–Cook (J–C) [59] constitutive model but excluding the effects of temperature and strain rate, the hardening law is expressed as

$$\sigma_y = A + B(\varepsilon^{\mathrm{p}})^n \tag{23}$$

where σ_y is the yield stress and $\varepsilon^{\mathrm{p}} = \int_0^t \sqrt{\frac{2}{3}D^{\mathrm{p}}_{ij}D^{\mathrm{p}}_{ij}}dt$ is the equivalent plastic strain. A is the initial yield stress, B is the hardening constant and n is a dimensionless hardening exponent. The initial yield strength A is 344 MPa by using the bias method (the bias value was taken as 0.2%) according to the standard ASTM E8M-04. And by the least squares method, B and n are determined as 654 Mpa and 0.61, respectively. Fig. 4(c) shows the fitted stress–strain curve as the red line. Density was selected as 7.85×10^{-9} tonne/mm^3 for dynamic simulation. All the material parameters used in the simulation are shown in Table 1.

3.3. FE simulation

Once the basic cell geometry has been designed, metamaterials can be created by tessellating these cells multiple times. The number of tessellations can vary based on the user's application. Sufficient tessellation is necessary to ensure uniform behavior and minimize the effects of boundary conditions. However, many unit cell designs must be tessellated 5–10 times before homogenization is met [60]. Although there are existing strategies for managing computational cost and improving efficiency, such as reduced-order models [61], adaptive mesh methods [62,63], high-performance linear algebra solvers [64,65], and parallel computing [66], this still makes designing, simulating, and optimizing the tessellated metamaterials computationally expensive. And this study focuses on demonstrating the ability of proposed methods to accomplish a strongly nonlinear design task. To reduce computational costs, a single basic cell is selected for FE simulation.

Fig. 5 illustrates the FE simulation process. The two-dimensional metamaterial with size 10 mm × 10 mm was positioned between two rigid plates shown in Fig. 5(a). Since initial peak crushing stress, plateau stress σ_m, and densification strain ϵ_m are key indicators for evaluating the effectiveness of energy absorbers in providing adequate protection, it is essential to simulate the sufficient compressive deformation behavior of metamaterials. In this study, a quasi-static compressive strain of up to 60% in the vertical direction was applied [12]. The deformation behavior of the two-dimensional metamaterial subjected to quasi-static compression was simulated using the ABAQUS [67] explicit software package. Under the assumption of plane strain conditions, the

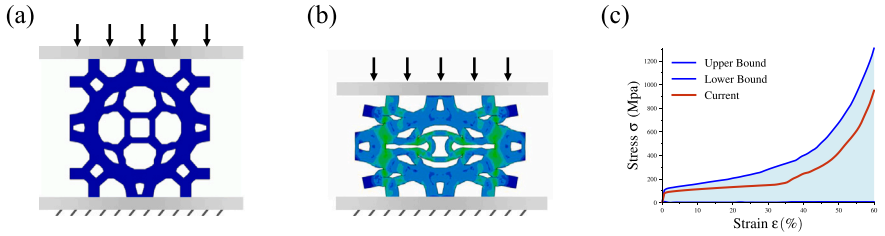


Fig. 5. A schematic of FE simulation. (a) load condition; (b) compression simulation; (c) stress–strain curve.

solid portion of the material is discretized using CPE6M elements. R2D2 elements are utilized for each rigid plate. The global seed size is set to 0.15 mm to ensure appropriate resolution throughout the simulation domain and mesh size independent. In order to avoid interpenetration, the interaction between the steel plate and the structure as well as the self-interaction within the structure are considered. For metal material, plastic yielding rather than elastic buckling is the primary mechanism contributing to the initial structural damage. Consequently, the simulation does not account for buckling-induced damage.

Fig. 5(b) shows the compression simulation of microstructure. Moreover, the stress values corresponding to uniformly sampled strain points were obtained through compression simulation. In Fig. 5(c), the red line represents the stress–strain curve corresponding to the microstructure shown in Fig. 5(a), and two blue lines represent the upper and lower bounds of all stress–strain curves.

3.4. Data analysis and process

3.4.1. Data visualization

Fig. 6 presents a detailed comparison of microstructures and their corresponding stress–strain curves. This figure highlights the significant variations in stress–strain behavior among the different microstructures. The stress–strain curve of the generated energy-absorbing metamaterial can be divided into three stages: the elastic period, the stable platform period, and the densification period. During the elastic period, the microstructure undergoes primarily elastic deformation, with stress increasing rapidly with strain. When the external force is removed, the material can return to its original shape. The stable platform period is characterized by plastic deformation, during which the energy absorbed increases significantly due to the occurrence of dislocations and slips within the material. This deformation is irreversible, leading to permanent changes in the material. During the densification period, the pores are compacted, leaving almost no remaining voids for further deformation. As a result, the tangent modulus increases, causing stress to rise rapidly.

As shown in Fig. 6, the first column presents the stress–strain curves of various microstructures, along with their corresponding volume fractions V within range of [0, 1]. From the first row, it can be observed that the size of the pores has a significant impact on the material's plastic platform period. Specifically, larger pores result in a longer plastic platform period, whereas smaller pores lead to a shorter period. Additionally, at the same strain level, the stress values generally increase with an increase in volume fraction. The second row demonstrates that different microstructures can significantly affect the stress–strain curves even with the same volume fraction. This highlights the crucial role of microstructure in energy absorption design. The third row shows that, in some cases, different microstructures can lead to nearly identical stress–strain curves. For example, cases 9 and 10, as well as cases 11 and 12, exhibit almost identical stress–strain curves despite having completely different microstructures. This indicates the complex relationship between microstructure and energy absorption curves, making the design of efficient energy-absorbing materials more challenging.

3.4.2. Stress–strain data postprocessing

Principal Component Analysis (PCA). We use PCA to reduce the dimensionality of stress data from uniform strain sampling. This process aims to reduce the complexity of the stress–strain data, leading to a better understanding of the structure and characteristics of the data. For d-dimensional stress corresponding to uniform strain sampling, PCA projects stress data in a d-dimensional space into a k-dimensional subspace, where k is constrained to be less than d. The k newly generated dimensions are denoted as principal components (PC). Each principal component is oriented to capture the maximum variance while excluding the variance already explained by its preceding components. Consequently, the first PC encapsulates the highest variance, with each successive PC accounting for progressively smaller variances.

To provide a proper normalization and give equal weights in the analysis phase of the original dataset, The first step of PCA analysis is subtracting the mean values from the stress dataset $\mathbf{Y} \in \mathbb{R}^{d\times n}$, where d represents the dimension of stress data, and n represents the number of samples

$$\bar{\boldsymbol{Y}}_j = \boldsymbol{Y}_j - E(\mathbf{Y}) \tag{24}$$

where $\bar{\boldsymbol{Y}}_j \in \mathbb{R}^d$ denotes jth normalization sample and $E(\mathbf{Y}) \in \mathbb{R}^d$ is the mean of $\mathbf{Y}$. Then, the covariance matrix is obtained as follows

$$C_{\bar{\mathbf{Y}}} = \mathrm{Cov}(\bar{\mathbf{Y}}, \bar{\mathbf{Y}}) = \frac{1}{n-1}\bar{\mathbf{Y}}\bar{\mathbf{Y}}^T \tag{25}$$

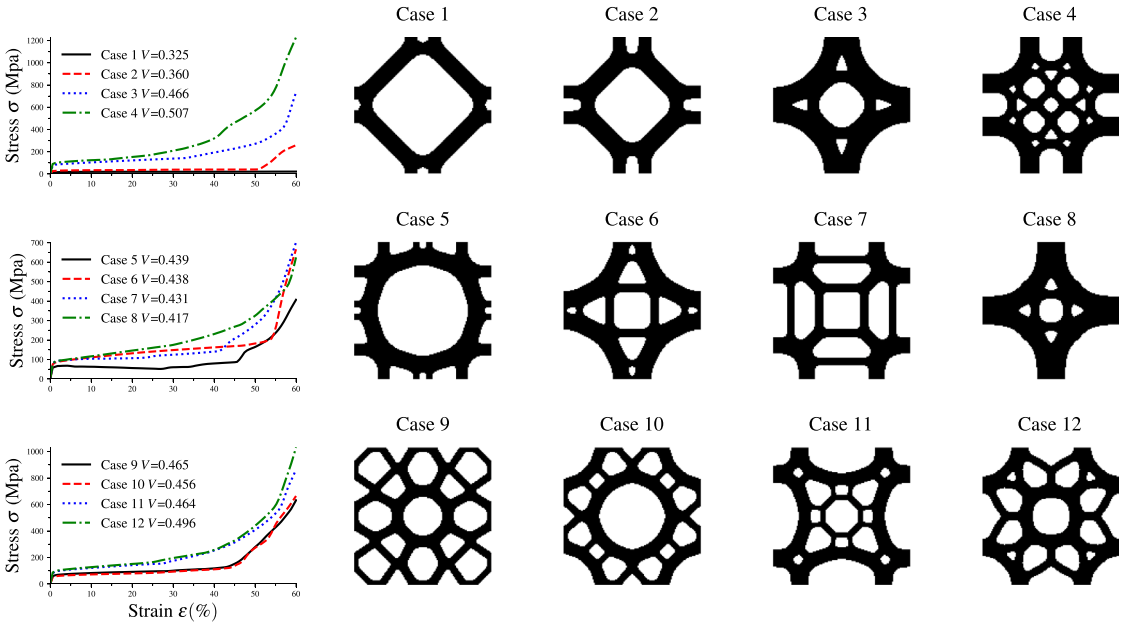


Fig. 6. Visualization of metamaterials and stress–strain curves.

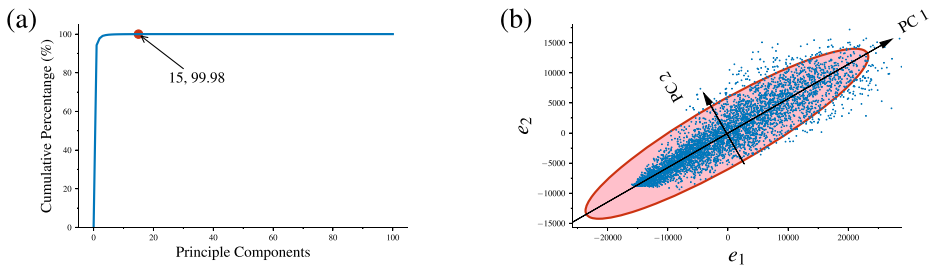


Fig. 7. PCA: (a) cumulative percentage of principal components; (b) the distribution of the stress data in the first and second principal component projections.

where $\bar{\mathbf{Y}}$ is normalized dataset by Eq. (24). Eigenvectors and eigenvalues of the covariance matrix can be obtained by singular value decomposition (SVD)

$$C_{\bar{\mathbf{Y}}} = \mathrm{Cov}(\bar{\mathbf{Y}}, \bar{\mathbf{Y}}) = \mathbf{U}\mathbf{D}^2\mathbf{U}^T \tag{26}$$

where $\mathbf{U}$ are the eigenvectors of $C_{\bar{\mathbf{Y}}}$, and the diagonal elements of $\mathbf{D}^2 = \begin{bmatrix}\lambda_1 & \lambda_2 & \cdots & \lambda_d\end{bmatrix}$ are the eigenvalues of $C_{\bar{\mathbf{Y}}}$. Finally, the stress dataset is transformed into a new axis with reduced dimensions with the eigenvectors obtained from Eq. (26)

$$\widetilde{\mathbf{Y}} = \mathbf{U}^T\mathbf{Y} \tag{27}$$

where $\widetilde{\mathbf{Y}} \in \mathbb{R}^{k\times n}$ denotes the reduced-dimension stress dataset such that $k < d$. The cumulative percentage (CP) of variance explained is calculated as follows

$$CP_m = \frac{\sum_{i=1}^{m}\lambda_i}{\sum_{i=1}^{d}\lambda_i} \times 100\% \tag{28}$$

where λ_i represents the variance contribution of the ith principal component. Using PCA, we can generate a cumulative distribution plot of PCs. As illustrated in Fig. 7(a), it becomes evident that the first fifteen principal components have the capacity to capture 99.98% of the variance. This signifies that, even with a significant reduction in dimensions, the majority of pivotal information inherent in the data remains preserved. Therefore, we decided to project the raw stress data into a space with only fifteen dimensions, making the data more compact while preserving the main data features. The distribution of the data in the first and second PCs projections shown in Fig. 7(b), demonstrates the distribution of the data in two dimensions after dimensionality reduction. The variance of the first PC is greater than that of the second PC. From Fig. 8, it can be observed that the reconstructed data closely matches the original curves after transforming the PCA-reduced data back into its original space.

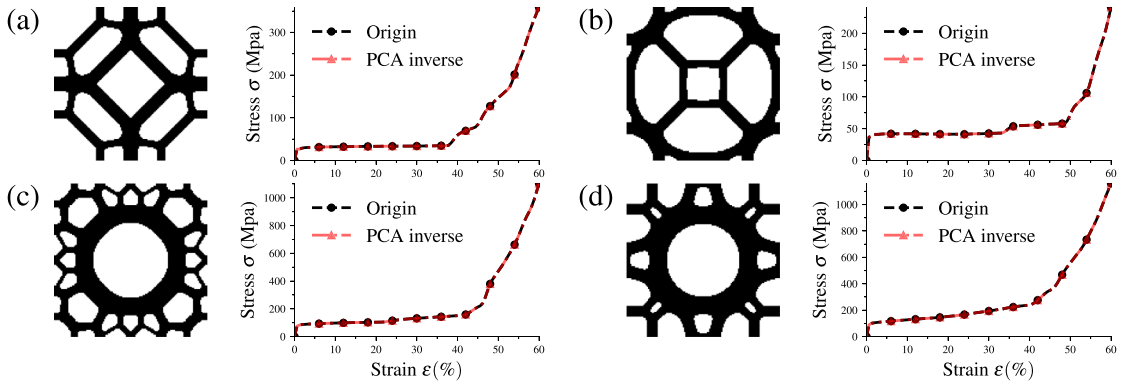


Fig. 8. Transform the PCA-reduced data back to its original space.

Normalization. Zero–one normalization is a common and important task in data preprocessing, and it can map the values of different features to a unified scale and eliminate the problem of model bias caused by differences between features, thereby improving the performance and stability of the machine learning model. The zero–one normalization is defined as follows:

$$y_i^{\sigma} = \frac{\widetilde{Y}_i - \widetilde{Y}_{i,\min}}{\widetilde{Y}_{i,\max} - \widetilde{Y}_{i,\min}} \tag{29}$$

where $\widetilde{Y}_i^{\sigma}$ is the ith value of reduced-dimension stress data, $\widetilde{Y}_{i,\max}^{\sigma}$ and $\widetilde{Y}_{i,\min}^{\sigma}$ are maximum and minimum of the ith values of reduced-dimension stress data, respectively. The normalized y_i^{σ} is within the range of $[0, 1]$.

3.4.3. Metamaterials postprocessing

Since our metamaterials design is based on pixels, the terms "microstructure" and "image" are interchangeable in this paper, as they convey the same information and meaning in our research. This interchange of terminology ensures consistency in our descriptions across various contexts and aids readers in better comprehending the subject matter under discussion. Due to the mirror symmetry in the generated microstructures in both the horizontal and vertical directions, to fully harness this mirror symmetry and reduce redundant data, we chose one-quarter of the original images, specifically microstructures of size 64 × 64 pixels for model training. By employing this approach, we can reduce data dimensionality while substantially preserving information integrity.

3.4.4. Partition training and testing sets

We partition the 30,000 microstructures and their stress–strain features into training and testing sets. In this partitioning process, 90% is used for training, while the remaining 10% is used for testing. This partitioning scheme is designed to maintain a balanced data set and allow us to take full advantage of the diversity and representation of the data in both the critical stages of training and testing.

3.5. Model architecture

In all experiments, we set $T = 1000$ so that the number of neural network evaluations needed during sampling matches previous work [37,47,68]. Furthermore, we have established the forward process variances as constants that increment linearly from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$. This setup ensures that both the reverse and forward processes exhibit a closely matched functional form.

To represent the reverse process, we employ a U-Net backbone similar to an unmasked PixelCNN++ [69,70] with group normalization throughout [71], as shown in Fig. 9. The U-Net consists of three main components: the encoder, decoder, and skip connections. The encoder compresses input images into a low-dimensional latent space through convolutional layers. The decoder takes the encoded input and maps it to the original resolution. The skip connections connect the encoder and decoder, creating a bottleneck in the latent space that emphasizes critical features, facilitates information flow in both directions, and reduces parameters.

Our models employ five different feature map resolutions, mapping images from 64 × 64 to 4 × 4. At each resolution level, there is a single convolutional residual block. To specify the diffusion time t, we incorporate the transformer sinusoidal position embedding [72] into each residual block. Model parameters are shared across time. To enhance the flexibility of the DiffMat as conditional microstructure generators, we used U-Net backbone with the cross-attention mechanism [30] for integrating mechanical information (such as stress–strain curves), which proves effective for learning attention-based models of various input modalities [50]. We use cross-attention at 32 × 32 and 4 × 4 feature map resolution. In order to embed mechanics information $\boldsymbol{y}^{\sigma}$, we introduce a domain specific encoder τ_θ that projects $\boldsymbol{y}^{\sigma}$ to an intermediate representation $\tau_\theta(\boldsymbol{y}^{\sigma}) \in \mathbb{R}^{d_\tau}$, which is then mapped to U-Net via a cross-attention layer implementing $\text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d}}\right) \cdot \boldsymbol{V}$, with [50]

$$\boldsymbol{Q} = \varphi_i(\boldsymbol{z}_t) \cdot \boldsymbol{W}_Q^{(i)} \quad \boldsymbol{K} = \tau_\theta(\boldsymbol{y}^{\sigma}) \cdot \boldsymbol{W}_K^{(i)} \quad \boldsymbol{V} = \tau_\theta(\boldsymbol{y}^{\sigma}) \cdot \boldsymbol{W}_V^{(i)} \tag{30}$$

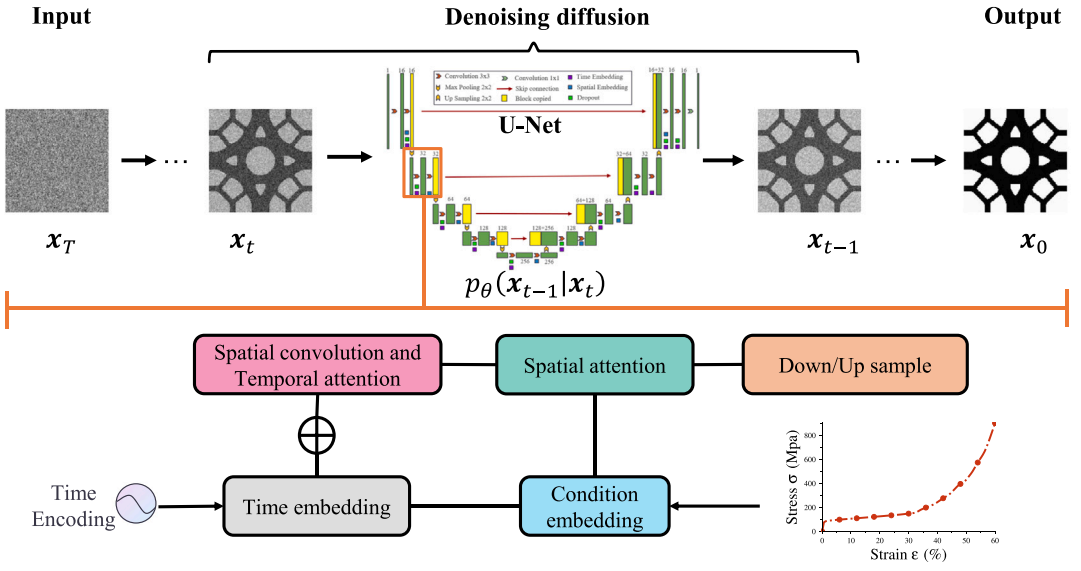


Fig. 9. Denoising diffusion model architecture.

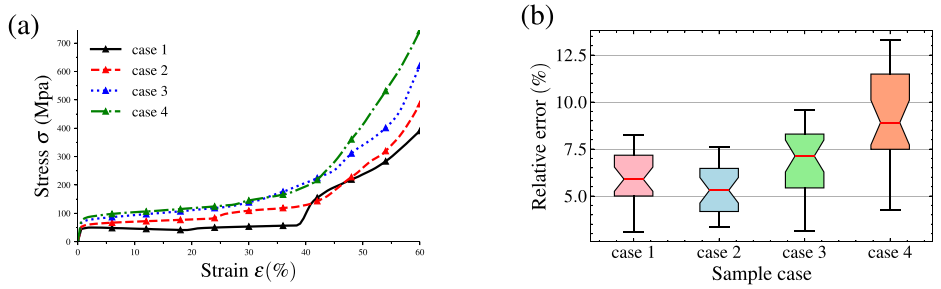


Fig. 10. Test cases: (a) test curves; (b) box plot of error d_{L_1} between the generated curves and the test curves.

Here, $\varphi_i(z_t) \in \mathbb{R}^{N\times d_\epsilon^i}$ denotes a (flattened) intermediate representation of the U-Net implementing ϵ_θ and $W_V^{(i)} \in \mathbb{R}^{d_\epsilon^i \times d}$, $W_Q^{(i)} \in \mathbb{R}^{d_\tau \times d}$ and $W_K^{(i)} \in \mathbb{R}^{d_\tau \times d}$ are learnable projection matrices [72].

Based on microstructure-mechanical condition dataset from Section 3.4.4, the training objective is expressed in Eq. (16). Our DiffMat model has 1.9 million parameters. We used Adam as optimizator with learning rate of 10^{-4} and implemented learning rate schedule, which decays the learning rate by 0.98 every 50 epochs. A total of 2000 epochs were performed according to Algorithm 1.

4. Examples

4.1. Test curve

Since DiffMat can learn the conditional probability distribution under given mechanical conditions, it can generate a set of microstructures with properties close to the target mechanical properties by sampling different initial noises $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. We selected four different stress–strain curves from the testing set as target curves for evaluating the test cases, shown in Fig. 10(a). For each curve, 50 noise sets were randomly sampled to generate microstructures according to Algorithm 2. Subsequently, the generated microstructures were imported into ABAQUS to obtain the corresponding stress–strain curves. To quantify the difference between the target curve and the other curve, the error d_{L_1} is defined as follows

$$d_{L_1}(Y^g, Y^t) = \frac{1}{d}\sum_{i=1}^{d}(|Y_i^g - Y_i^t|/Y_i^t) \times 100\% \tag{31}$$

where $Y^t \in \mathbb{R}^d$ denotes the target stress–strain curve and $Y^g \in \mathbb{R}^d$ denotes the other stress–strain curve.

Fig. 10(b) presents a box plot of the error d_{L_1} between the curves of generated microstructures (referred to as generated curves) and the target curves. For these four target curves, the median error d_{L_1} between the generated curves and the target curves is

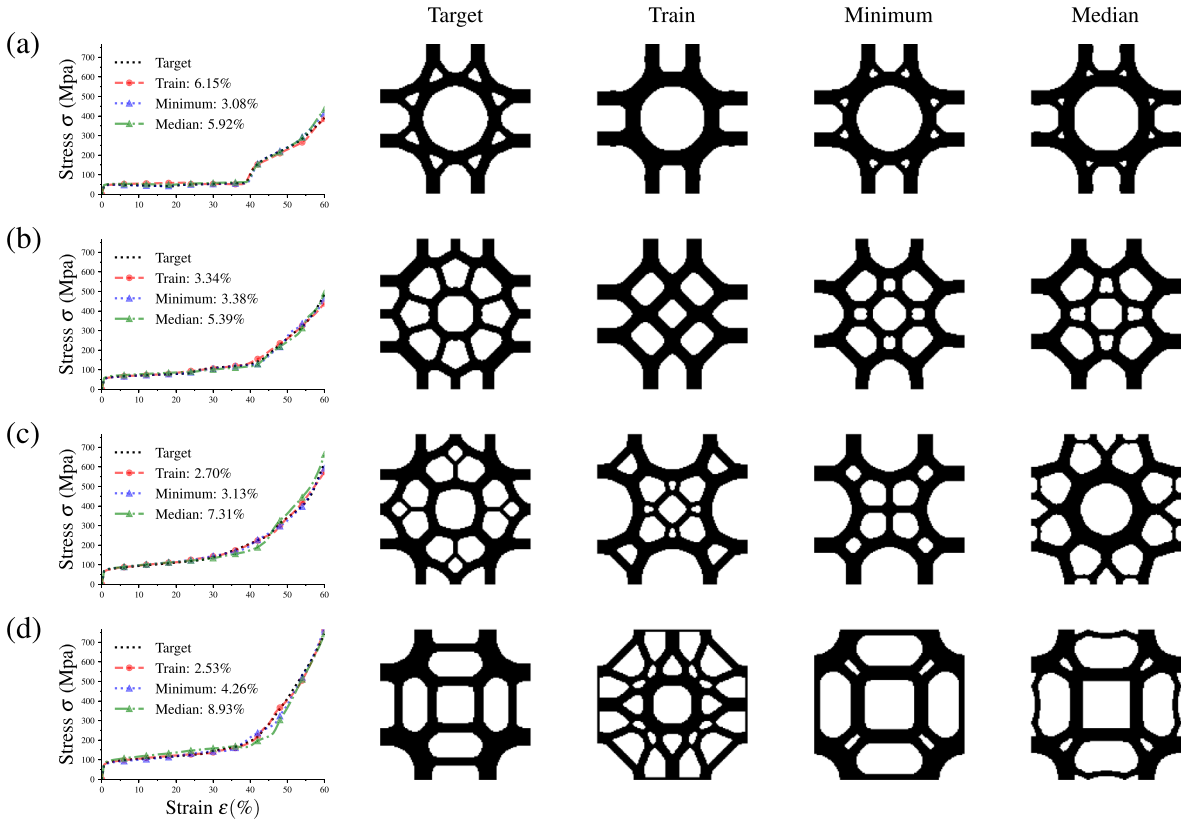


Fig. 11. Generated microstructures and their stress–strain curves for test cases: (a) case 1; (b) case 2; (c) case 3; (d) case 4.

less than 9% for all cases. This indicates that DiffMat can accurately generate microstructures with the given stress–strain curve properties.

Furthermore, Fig. 11 display the microstructures and their stress–strain curves corresponding to the minimum and median values of d_{L_1} of target curves. Each row corresponds to the generated results for each case. In the first column, the target curves are represented by the black line. The generated curves corresponding to the minimum d_{L_1} are shown in blue, the generated curve corresponding to the median d_{L_1} are shown in green, and the red line represents the curves from the training dataset (referred to as the training curve) with the minimum d_{L_1} relative to the target curve. The second column shows the microstructures corresponding to the target curves. The third column shows the microstructures corresponding to the minimum d_{L_1} of the training curves. The fourth column shows the microstructures corresponding to the minimum d_{L_1} of the generated curves. The fifth column shows the microstructures corresponding to the median d_{L_1} of the generated curves.

From the first column of Fig. 11, it is evident that the minimum error d_{L_1} between the target curves and the generated curves is less than 4.5%. Additionally, the generated microstructures in the fourth and fifth columns differ significantly from the target microstructures in the second column. This demonstrates the capability of DiffMat to generate distinctly different microstructures under the given target curve. This notable feature enhances the flexibility of microstructures design and aids in discovering microstructures that meet the desired mechanical performance criteria.

In Fig. 11(a), the first column shows that the minimum error d_{L_1} of the training curve is greater than the minimum error d_{L_1} of the generated curve. In Fig. 11(b), (c), (d), the first column reveals that the minimum error d_{L_1} of the generated curves is close to the minimum error d_{L_1} of the training curves. Simultaneously, there are noticeable differences between the generated microstructures in the fourth and fifth columns and the training microstructures in the third column. This further demonstrates DiffMat's strong generative capability, enabling it to generate microstructures beyond the training set and allowing for the discovery of superior solutions within the design space.

By comparing the fourth and fifth columns in Fig. 11, it is evident that while the curves corresponding to the generated microstructures are similar, their microstructures differ significantly. This observation underscores that DiffMat learns the conditional distribution of microstructures given specific mechanical properties, achieving a one-to-many mapping from properties to geometry. It showcases the flexibility and robust power of DiffMat in generating various geometric microstructures with the same mechanical properties.

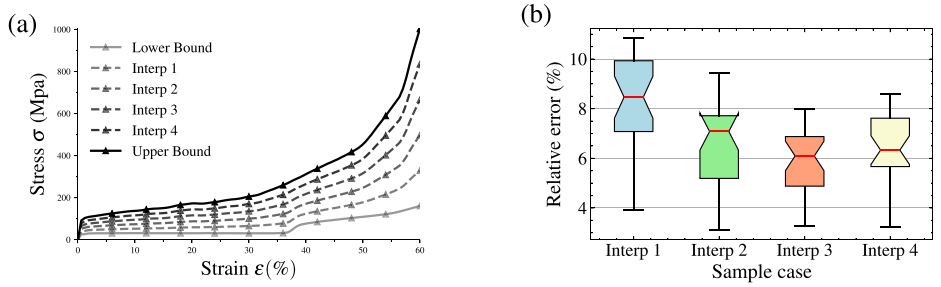


Fig. 12. Interpolation case 1: (a) interpolation curves; (b) box plot of error d_{L_1} between the generated curves and the interpolated curves.

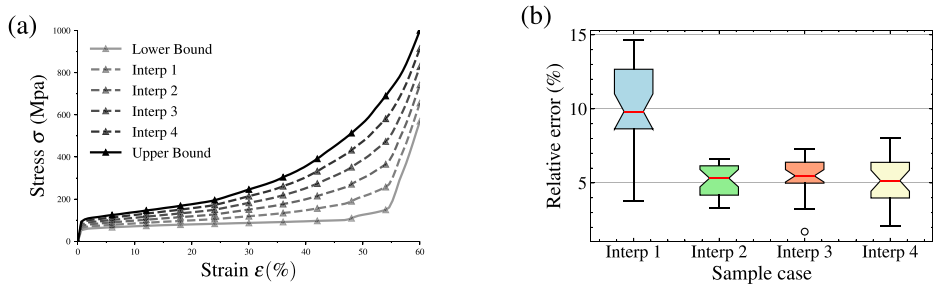


Fig. 13. Interpolation case 2: (a) interpolation curves; (b) box plot of error d_{L_1} between the generated curves and the interpolated curves.

4.2. Interpolation curve

Since the curves in the testing set might be similar to those in the training set, we performed an interpolation process to evaluate the generative capability of DiffMat further and to gain deeper insights into its generalization. Specifically, we selected two stress–strain curves from the training set and obtained a set of interpolated target curves through linear interpolation between these two curves. In total, two interpolation cases for condition generation are investigated, with each case containing four interpolated curves as target curves.

In Figs. 12(a) and 13(a), the solid lines represent the two curves selected from the training set, and the dashed lines represent the interpolated curves, i.e., target curves. Figs. 12(b) and 13(b) show box plots of the error d_{L_1} between the generated curves and the interpolated curves for the two interpolation cases, respectively. For all four interpolated curves of each interpolation case, the median error d_{L_1} between the generated curves and the interpolated curves is less than 9%. This demonstrates the exceptional generative capability and precision of DiffMat, producing microstructures with properties that match the target curves under complex interpolation cases.

Furthermore, Figs. 14 and 15 display the microstructures and their stress–strain curves corresponding to the minimum and median values of d_{L_1} for each interpolation case. Each row corresponds to generated results for each interpolation curve. In the first column, the black line represents the target curves, i.e., interpolated curves, the blue line represents the generated curves with the minimum d_{L_1}, the green line represents the generated curves with the median d_{L_1}, and the red line represents the training curves with the minimum d_{L_1}. The second column shows the microstructures corresponding to the minimum d_{L_1} of the training curves. The third column shows the microstructures corresponding to the minimum d_{L_1} of the generated curves. The fourth column shows the microstructures corresponding to the median d_{L_1} of the generated curves.

By comparing the generated microstructures in Figs. 14 and 15, it is evident that adjacent microstructures exhibit smooth, gradual transitions rather than abrupt or discontinuous changes as the interpolation curves change. This highlights the ability of DiffMat to maintain continuity and consistency as the interpolated target stress–strain curves change, reflecting a deep understanding of the complex relationship between microstructures and mechanical properties and demonstrating the reliability and stability of conditional generation. Additionally, the smooth transitions produced by DiffMat facilitate the exploration of a broader design space, potentially leading to the discovery of new metamaterial topologies with superior performance or unique properties and opening up new possibilities and innovations in the field of metamaterial design.

The first column in Figs. 14 and 15 shows that, in most cases, the minimum error d_{L_1} in the training curves is greater than the minimum error d_{L_1} in the generated curves. The corresponding microstructures in the second and third columns exhibit significant differences. This finding highlights DiffMat's exceptional generative capabilities. By learning and understanding the characteristics of the target curves, DiffMat can generate suitable microstructures for these conditions, achieving a precise mapping from mechanical properties to microstructural geometry. This ability allows DiffMat to surpass the limitations of the training set, generating novel microstructures and demonstrating immense potential in diversity and innovation.

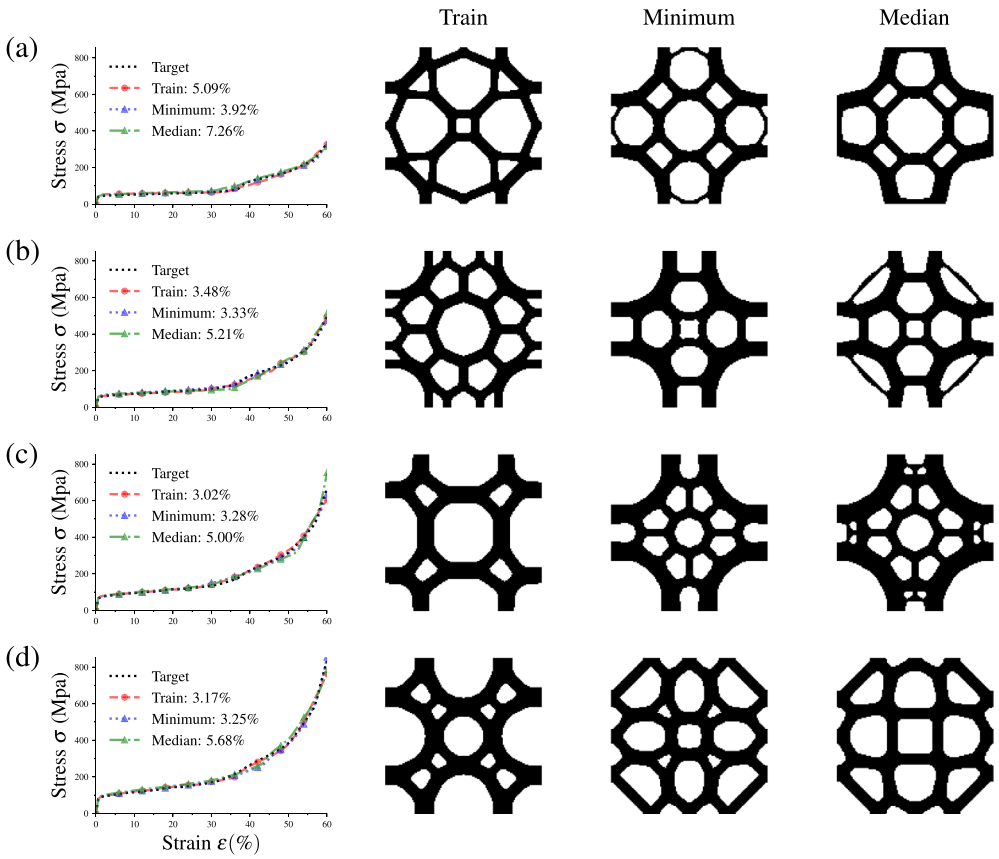


Fig. 14. Generated microstructures and their stress–strain curves of interpolation case 1: (a) interpolated curve 1; (b) interpolated curve 2; (c) interpolated curve 3; (d) interpolated curve 4. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.3. Experimental verification

This section conducted a series of experiments to verify the performance and accuracy of DiffMat. Firstly, the experimental samples of metamaterials were prepared by machining. Next, quasi-static compression experiments were carried out using a tensile testing machine, with the compression constant loading rate set at 1 mm/min and the loading conditions shown in Fig. 16(a). The stress–strain curves of the microstructures were obtained as shown in Fig. 16(b). Digital image correlation (DIC) technology is used to measure the deformation of the metamaterials more accurately. The acquisition equipment used in the experiment is shown in Fig. 16(c), which can effectively record and store the data generated during the experiment, including important information such as loading conditions, deformation, and mechanical response of the metamaterial. We chose the generated microstructures with minimum error d_{L_1} of interpolation case 1 for experiments. The experimental results are shown in Fig. 17.

Fig. 17(a) presents a comparison between the experimental and simulation results for interpolated curve 1. The stress–strain curves of the metamaterial are shown on the left. The black solid curve represents the target curve, the red dotted curve represents the simulated curve of the generated metamaterial, and the blue dash-dot curve represents the experimental curve of the generated metamaterial. The error d_{L_1} between the experimental curve and the target design curve is 15.2%. The compression process is depicted on the right side (I–IV), with the upper section showing the simulation contour plots and the lower section showing the experimental deformation images. As the strain increases, the internal tiny pores close first (Fig. 17(a)II, III). When the strain reaches 45% (Fig. 17(a)IV), the central pore is almost completely closed. With further strain increase, the upper and lower rods contact the compression plate (Fig. 17(a)V), leading to a rapid rise in the tangent modulus. Additionally, it is observed that the plastic deformation areas in the experimental deformation diagrams align with the regions of maximum stress in the simulation contour plots. This consistency indicates the accuracy of both the simulation and experimental results.

Fig. 17(b) shows the comparison between the experimental and simulated results of interpolated curve 2, with an error d_{L_1} of 15.3% between the experimental curve and the target design curve. As the strain increases, the internal voids compress (see Fig. 17(b)II). Between 40%–55% strain, the metamaterial experiences instability (see Fig. 17(b)III, IV), causing the upper and lower struts to prematurely contact the pressure plate, resulting in the experimental curve in this segment being higher than the simulated curve. As the strain continues to increase, at 55% strain, the central void gradually closes, and the tangent modulus rapidly increases (see Fig. 17(b)V).

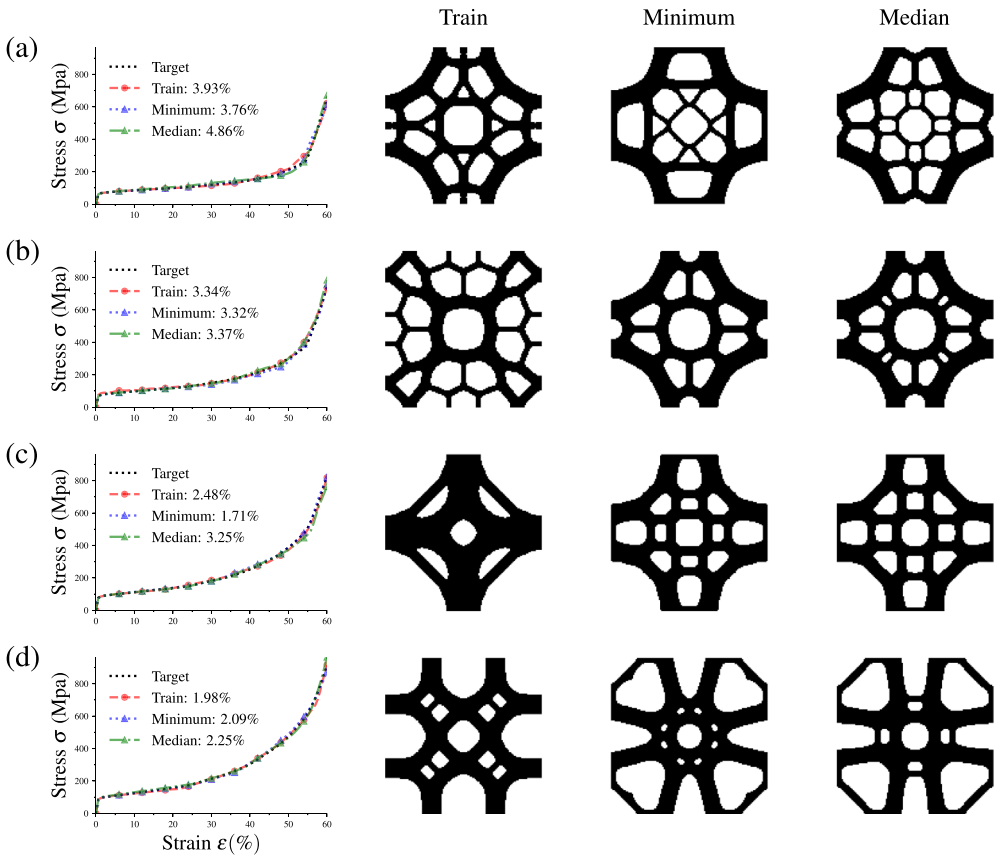


Fig. 15. Generated microstructures and their stress–strain curves of interpolation case 2: (a) interpolated curve 1; (b) interpolated curve 2; (c) interpolated curve 3; (d) interpolated curve 4. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

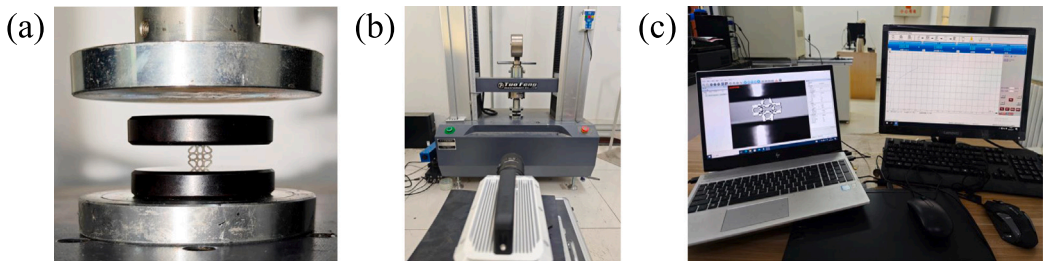


Fig. 16. Metamaterial compression experiment: (a) compression condition ; (b) digital image correlation; (c) acquisition equipment.

Fig. 17(c) presents the comparison of the experimental and simulated results for interpolated curve 3, with an error d_{L_1} of 13.5% between the experimental curve and the target design curve. As the strain increases, the internal voids compress (see Fig. 17(c)II, III), exhibiting a trend towards densification, with clear shear bands appearing within the internal struts. It is also observed that around 40% of strain, there are fluctuations attributed to slight instability of the metamaterial due to factors such as machining errors and loading conditions (see Fig. 17(c)IV), resulting in discrepancies with the simulated curve. As the strain continues to increase, at 58% strain, the internal voids completely close, and the tangent modulus rapidly increases (see Fig. 17(c)V).

Fig. 17(d) shows the comparison of the experimental and simulated results for interpolated curve 4, with an error d_{L_1} of 11.2% between the experimental curve and the target design curve. As the strain increases, the internal small voids rapidly close (see Fig. 17(d)II, III), with clear shear bands appearing near the closed voids. With further strain increase, larger voids gradually close (see Fig. 17(d)IV), leading to significant plastic deformation within the internal structures. At 55% strain, due to the nearly closed voids (see Fig. 17(d)V), the contact area between the material and the pressure plate increases, rapidly increasing the tangent modulus.

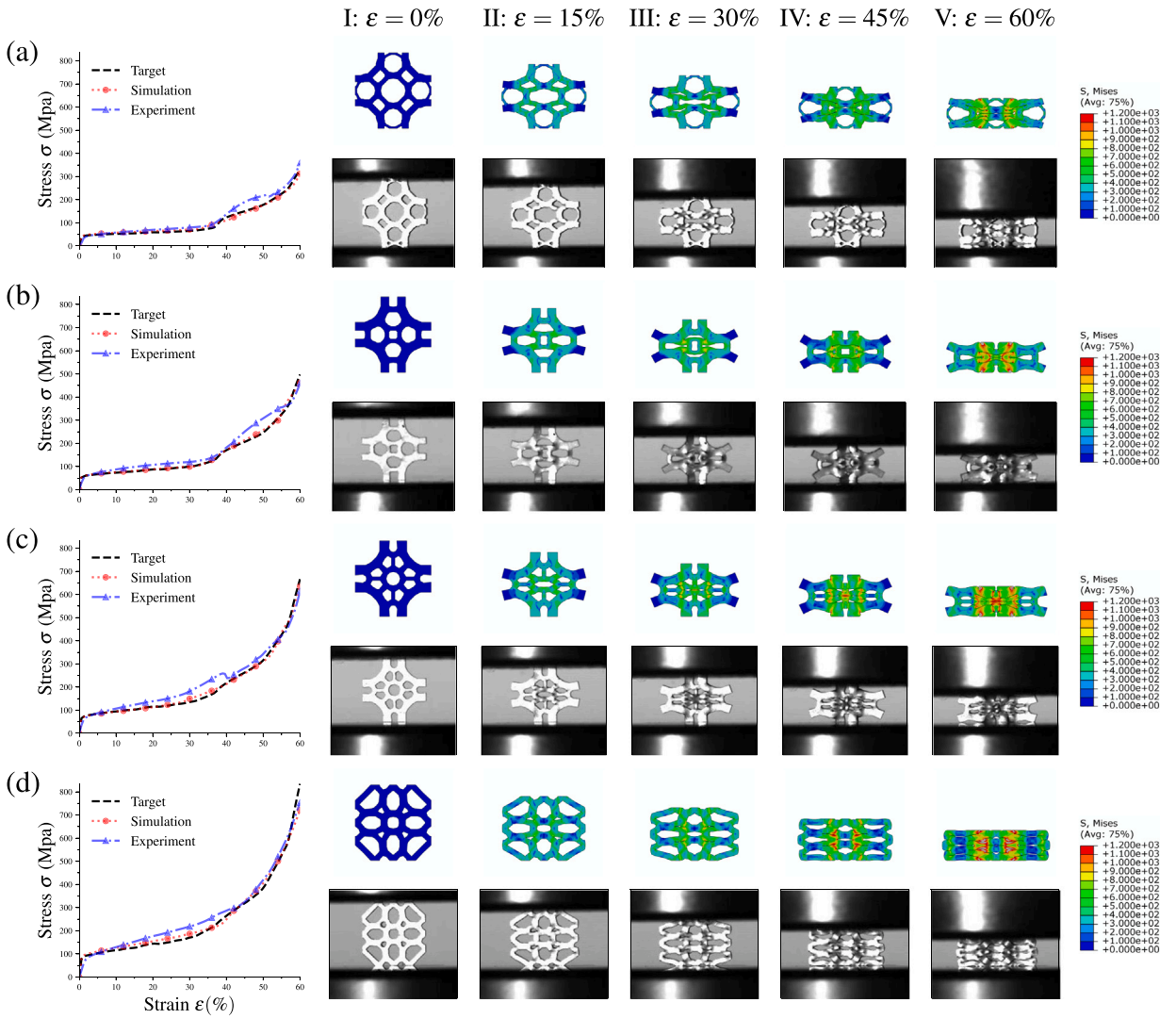


Fig. 17. The compression experiment verification of energy absorbing metamaterials. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5. Discussion

Through numerical examples and experimental validation, DiffMat has shown an exceptional ability to learn the conditional distribution of microstructures given specific mechanical properties. This method enables a one-to-many mapping from mechanical properties to geometric structures, allowing for the generation of microstructures that exceed the range of the training set under a specified target stress–strain curve. Moreover, DiffMat maintains continuity and consistency in the generated microstructures when interpolating variations in target stress–strain curves. This capability highlights DiffMat's broad applicability and reliability in material and structure design and its outstanding performance in meeting complex design requirements.

The effectiveness of this framework can be attributed to the following key factors:

1. High-quality training data: Carefully designed experiments and the selection of appropriate material constitutive models were employed to ensure that the simulation accurately reflects the mechanical behavior of materials under real working conditions. High-quality training data serves as the foundation for ensuring the accuracy and reliability of this framework.
2. Flexible topology representation: The topology space was divided into a series of pixels, ensuring the flexibility of the topology representation and the breadth of the design space. This flexibility enables DiffMat to adapt to diverse design requirements and perform well in complex mechanical environments.
3. Efficient feature extraction: PCA was utilized for dimensionality reduction of the stress–strain data, which simplified the complexity of the training data and enhanced the understanding of its underlying structure and features. Additionally,

zero–one normalization was applied to minimize discrepancies between features. During model training, using PCA and normalization was found to substantially ease the neural network training process while boosting the model's performance and stability.

4. Advanced network model architecture: This study employs a UNet model with attention mechanism, where specially designed skip connections form bottlenecks in the latent space, emphasizing the capture of key features, promoting bidirectional information flow, and reducing model parameters. Meanwhile, the cross-attention mechanism helps efficiently integrate mechanical information, enhancing the overall performance of the model.

Although designed in this paper only for large deformation elastoplastic materials, the method as a data-driven approach can be extended to nonlinear behaviors such as visco-plasticity, damage, etc., and to take into account other physical properties (e.g., thermal, optical, acoustic, or combinatorial). To further enhance the generality of the DiffMat framework and advance the development of data-driven design paradigms, several suggestions for improvement and expansion are proposed:

1. High-quality and diverse training data: High-quality training data not only needs to be obtained through precise simulations but also must cover the design space. Noise, incompleteness, or biases in the training data can directly impact the model's learning capability and reduce design accuracy. Future work should focus on exploring diversity within the attribute space [73], such as database expansion, Pareto frontier discovery [74], and optimal diversity coverage to better support diverse applications.
2. Efficient topology representation: Choosing an appropriate topology representation is crucial for specific problems. The representation should be closely related to the research problem and have high representational efficiency to reduce computational resource consumption and improve design accuracy. Additionally, it should be flexible enough to allow for a broad exploration space, fully exploring possible structural configurations during the design process.
3. Data analysis and mining based on prior knowledge: Effective feature representation can not only better present key features but also enhance the overall performance of the model. For specific domains, particular attention should be paid to integrating domain knowledge and physical mechanisms [75] to ensure the effectiveness and accuracy of data processing.
4. Efficient and robust network model architecture: With the rapid development of AI technologies, more advanced model architectures for complex problems will continue to emerge. DiffMat can leverage these new technologies to improve its architecture, making it more adaptive and flexible in handling complex design tasks. Considering the response variations under different experimental conditions and possible uncertainties and errors, building a more efficient and robust multi-fidelity model architecture [76] is particularly important to achieving broader engineering applications.

6. Conclusion

The present study introduces a novel framework for designing energy-absorbing metamaterials by utilizing an inverse design method based on a diffusion model. This method, called DiffMat, aims to create metamaterials with precise energy-absorbing curves. By analyzing the target curves, DiffMat captures the relationship between mechanical properties and geometry, enabling the framework to establish a one-to-many mapping and generate a diverse range of microstructures that exhibit the desired mechanical properties, allowing for customizing energy-absorbing metamaterials. The validity and effectiveness of DiffMat are successfully demonstrated through numerical simulations and experiments. The successful implementation of the inverse design method based on the diffusion model provides a powerful tool for designing energy-absorbing metamaterials. It contributes to the broader field of materials science and engineering. This breakthrough can potentially revolutionize the development of energy-absorbing materials and structures, leading to enhanced safety and performance in various industrial applications.

CRediT authorship contribution statement

Haoyu Wang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation. **Zongliang Du:** Writing – review & editing, Supervision, Formal analysis. **Fuyong Feng:** Supervision, Funding acquisition. **Zhong Kang:** Supervision, Funding acquisition. **Shan Tang:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Xu Guo:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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